

Video Communication & Collaboration Platform

MINIMUM VIABLE PRODUCT (MVP) SPECIFICATION

| PROJECT OVERVIEW

The Video Communication Platform is an enterprise-ready virtual collaboration solution built to host interactive meetings. It delivers real-time high-definition video, stable audio transmission, and shared workspace tooling for distributed teams.

| MAIN OBJECTIVE

The main objective of this platform is to provide organizations with a highly reliable, secure environment for holding instant or scheduled virtual meetings accompanied by multi-party collaboration features.

| USER ROLES

The MVP architecture focuses on three explicit participant configurations:

- **Administrators:** Organization-level usage audits, license configuration, and broad room compliance.
- **Meeting Hosts:** Room initialization, calendar scheduling, participant moderation, and presentation controls.
- **Participants:** Join access via web link, audio/video toggling, text chat, and interactive tool utilization.

| CORE MVP FEATURES

Meeting Management

Enables authorized users to instantly initiate video calls, schedule future occurrences, generate meeting invitations, and configure basic meeting templates.

Video Conferencing Engine

Delivers multi-participant HD video feeds supporting grid and speaker views, alongside essential client-side camera and mic controls.

Screen Sharing

Allows hosts and designated presenters to share their entire display, a specific application window, or a browser tab in real time.

In-Meeting Collaboration

Features a shared interactive whiteboard, collaborative session notes, live text chat, and quick attachment/file sharing capabilities.

| TECHNOLOGY STACK

- **Frontend:** React.js
- **Backend:** Node.js, Express.js
- **Database:** PostgreSQL
- **Additional Tools:** WebRTC (Video/Audio streaming), Socket.IO (Signaling & Chat), Redis

| CONCLUSION

This Video Communication MVP establishes a modern, low-latency framework for secure virtual collaboration, meeting the fundamental communication demands of a scaling workforce.